Seth Ricks

ECEN 470
10/21/2026

PSO6:
5.1(a,b,c-i, c-ii, c-iii),
5.13(a-use MATLAB)(b)(c)(d)

1/2

(5.1) Set up the listed characteristic equations in the form suited to Evans root-locus method. Give $L(s)$, $a(s)$, and $b(s)$ and the parameter K in terms of the original parameters in each case. Be sure to select K so $a(s)$ and $b(s)$ are monic in each case, and the degree of $b(s)$ is not greater than that of $a(s)$.

a.) $s + (1/\tau) = 0$ versus parameter τ

b.) $s^2 + cs + c + 1 = 0$ versus parameter c

c.) $(s+c)^3 + A(Ts+1) = 0$

i.) versus parameter A

ii.) " " T

iii.) " " c, if possible. Say why you can or cannot. Can a plot of roots be drawn versus c for given constant values of A and T by any means at all?

---

a.) $1 + K\frac{b(s)}{a(s)} = 0 \qquad 1 + \frac{1/\tau}{s} = 0, \quad 1 + \frac{1}{\tau}\frac{1}{s} = 0$

$K = \frac{1}{\tau}; \; b(s) = 1; \; a(s) = s; \; L(s) = \frac{b(s)}{a(s)} = \frac{1}{s}$

b.) $s^2 + cs + c + 1 = 0$

$s^2 + c(s+1) + 1 = 0$

$c(s+1) = -s^2 - 1$

$\frac{c(s+1)}{-s^2-1} = 1$

$\frac{c(s+1)}{-s^2-1} - 1 = 0$

$-\frac{c(s+1)}{s^2+1} - 1 = 0$

$1 + c\frac{(s+1)}{s^2+1} = 0$

$K = c; \; b(s) = s+1; \; a(s) = s^2+1; \; L(s) = \frac{s+1}{s^2+1}$

c.) $(s+c)^3 + A(Ts+1) = 0$

i.) versus A: $1 + AT\frac{(s+1/T)}{(s+c)^3} = 0$

$K = AT; \; b(s) = s + \frac{1}{T}; \; a(s) = (s+c)^3; \; L(s) = \frac{s+1/T}{(s+c)^3}$

ii.) versus T: $(s+c)^3 + ATs + A = 0$

$(s+c)^3 + A + (As)T = 0$

$1 + AT\frac{s}{(s+c)^3 + A} = 0$

$K = AT; \; b(s) = s; \; a(s) = (s+c)^3 + A; \; L(s) = \frac{s}{(s+c)^3 + A}$

iii.) versus c: NOT POSSIBLE: can't isolate c w/ c, c^2, c^3

Could solve numerically.

* See MATLAB

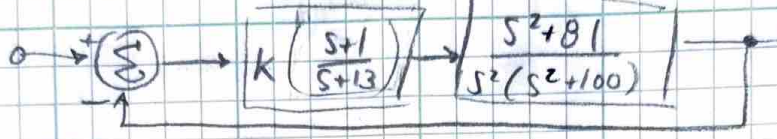
6.13 For system below, [2/2]

a.) Find locus of closed-loop roots w/ respect to K (use MATLAB)

b.) Is there val of K that causes all roots to have damping ratio greater than 0.5?

c.) Find values of K that yield closed-loop poles with damping ratio $\zeta = 0.707$

d.) Use MATLAB to plot response of the resulting design to a reference step



a.) Char eq: $1 + K\frac{(S+1)(S^2+81)}{S^2(S+13)(S^2+100)} = 0$ → see MATLAB for plot

b.) $\zeta = \sin\phi$

$\phi = \sin^{-1}\zeta = \sin^{-1}(0.5) = 30°$

$\phi \geq 30°$

There is NOT a value where $\zeta > 0.5$. This is because there are loci that don't cross that on the graph.

c.) $b(s) = (s+1)(s^2+81)$

$a(s) = s^2(s+13)(s^2+100)$

$1 + K\frac{b(s)}{a(s)} = 0$

dominant poles determined by magnitude and rate of decay.
↳ use graph

↳ $K \approx 88.2$ (w/ graph)

d.) See MATLAB